SF2940 Probability: Exam

Fall 2024

Date: 18. October 2024, 14:00-18:00.
Examiner: Kevin Schnelli, SCI MAT. Phone: 72 02
Aids: One page hand written notes. No other aids are allowed. Formulary with distributions is included after the problem statement.

General instructions: Your solutions and answers should be legible, easy to follow and not lack in explanations. For full points a correct and well-presented reasoning is required.

Exam design: This exam consists of four independent problems. Each problem gives at most 5 points.

Problem 1.

a.) Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a probability space. Consider two events $A, B \in \mathcal{A}$. Prove that

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)\,.$$

. . . . . (2points)

b.) Let X be a random variable with characteristic function $\varphi_X(t)$. Show that

$$(\varphi_X(t))^2$$

is also a characteristic function. . . . . (1point)

c.) Let (X_i), $i = 1, 2, 3, \ldots$ be a sequence of i.i.d. random variables. For fixed $x \in \mathbb{R}$, set

$$\widehat{F}_n(x) := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}(X_i \leq x)\,, \qquad n = 1, 2, 3, \ldots\,,$$

where

$$\mathbb{1}(X_i \leq x) = \begin{cases} 1\,, & \text{if} \quad X_i \leq x\,, \\ 0\,, & \text{if} \quad X_i > x\,. \end{cases}$$

Show that $\widehat{F}_n(x)$ converges in mean-square, as $n \to \infty$, and determine the limit.
. . . . . (2points).

Problem 2. Let $\mathbf{X} = (X_1, X_2, X_3)'$ be a three-dimensional Gaussian vector with mean vector identically equal to zero and covariance matrix

$$\Lambda_{\mathbf{X}} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{pmatrix} .$$

Set now $Y_1 := X_1 + X_3$ and $Y_2 := X_2$.

a.) Compute explicitly step by step the conditional density function of Y_2 given that $Y_1 = y_1$.

..... (2points)

b.) Compute the probability density function of the random variable

$$U := \frac{X_1}{X_2} .$$

..... (3points)

Problem 3. Let (X_i), $i = 1, 2, 3, \ldots$, be independent and identically distributed random variables whose probability mass functions are given by

$$p_{X_i}(k) = \frac{1}{\mathrm{e}^m - 1} \frac{m^k}{k!} , \qquad k = 1, 2, 3, \ldots ,$$

with $m > 0$ a fixed real number.

a.) Determine the probability generating function of every X_i, $i = 1, 2, 3, \ldots$. .(2points)

c.) Let n be natural number. Let $N \in \mathrm{Bin}(n, 1 - \mathrm{e}^{-m})$, and assume that N is independent of $X_1, X_2, \ldots$ above. Set

$$S_N := X_1 + X_2 + \ldots + X_N , \qquad S_N := 0 \quad \text{if} \quad N = 0 .$$

Determine the distribution of S_N ..... (3points)

Problem 4. Let the random variable X have Gamma distribution $\Gamma(s,1)$, with $s > 0$. Assume that

$$Y|X = x \in \mathrm{Po}(x)\,,$$

that is, given $X = x$, the random variable Y is Poisson distributed with parameter $x > 0$.

a.) Determine the characteristic function of Y. . . . . (2points)

b.) Show that

$$\frac{Y - \mathbb{E}[Y]}{\sqrt{\mathrm{Var}(Y)}}$$

converges in distribution as $s \to \infty$ and determine the limit. . . . . (3points)

Hint: The characteristic function of X is

$$\varphi_X(t) = \frac{1}{(1 - \mathrm{i}t)^s}\,.$$

Moreover,

$$\ln(1 - z) = -z - \frac{z^2}{2} + o(z^2)\,,$$

as $|z| \searrow 0$.

Good luck!

Distribution	Notation	Probability function	$\varphi_X(t)$
One point	$\delta(a)$, $a \in \mathbb{R}$	$p(a) = 1$	$\mathrm{e}^{\mathrm{i}ta}$
Bernoulli	$\mathrm{Be}(p)$; $0 \le p \le 1$	$p(0) = q$, $p(1) = p$; $q = 1 - p$	$q + p\mathrm{e}^{\mathrm{i}t}$
Binomial	$\mathrm{Bin}(n, p)$, $n = 1, 2, \ldots$; $0 \le p \le 1$	$p(k) = \binom{n}{k} p^k q^{n-k}$, $k = 0, 1, 2, \ldots, n$; $q = 1 - p$	$(q + p\mathrm{e}^{\mathrm{i}t})^n$
Geometric	$\mathrm{Ge}(p)$, $0 \le p \le 1$	$p(k) = pq^k$, $k = 0, 1, 2, \ldots$; $q = 1 - p$	$\frac{p}{1-q\mathrm{e}^{\mathrm{i}t}}$
Poisson	$\mathrm{Po}(\lambda)$, $\lambda > 0$	$p(k) = \mathrm{e}^{-\lambda}\frac{\lambda^k}{k!}$, $k = 0, 1, 2, \ldots$	$\mathrm{e}^{\lambda(\mathrm{e}^{\mathrm{i}t}-1)}$

Table 1. Discrete distributions

Distribution	Notation	Density	$\varphi_X(t)$
Uniform	$\mathrm{U}(a, b)$, $a < b$	$f(x) = \frac{1}{b-a}$, $a < x < b$	$\frac{\mathrm{e}^{\mathrm{i}tb}-\mathrm{e}^{\mathrm{i}ta}}{\mathrm{i}t(b-a)}$
Exponential	$\mathrm{Exp}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{\lambda}\mathrm{e}^{-x/\lambda}$, $x > 0$	$\frac{1}{1-\lambda\mathrm{i}t}$
Laplace	$\mathrm{L}(\lambda)$, $\lambda > 0$	$f(x) = \frac{1}{2\lambda}\mathrm{e}^{-\lvert x\rvert/\lambda}$, $x \in \mathbb{R}$	$\frac{1}{1+\lambda^2t^2}$
Beta	$\beta(r, s)$, $r, s > 0$	$f(x) = \frac{\Gamma(r+s)}{\Gamma(r)\Gamma(s)}x^{r-1}(1-x)^{s-1}$, $0 < x < 1$	$*$
Normal	$\mathrm{N}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(x-\mu)^2/\sigma^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}\mu t-\frac{1}{2}\sigma^2t^2}$
Standard normal	$\mathrm{N}(0, 1)$	$f(x) = \frac{1}{\sqrt{2\pi}}\mathrm{e}^{-\frac{1}{2}x^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{-\frac{1}{2}t^2}$
Log-normal	$\mathrm{LN}(\mu, \sigma^2)$, $\mu \in \mathbb{R}$, $\sigma > 0$	$f(x) = \frac{1}{x\sqrt{2\pi\sigma^2}}\mathrm{e}^{-\frac{1}{2}(\log x-\mu)^2/\sigma^2}$, $x > 0$	$*$
Cauchy	$\mathrm{C}(m, \eta)$, $\eta > 0$, $m \in \mathbb{R}$	$f(x) = \frac{1}{\pi}\frac{\eta}{(x-m)^2+\eta^2}$, $x \in \mathbb{R}$	$\mathrm{e}^{\mathrm{i}mt-\eta\lvert t\rvert}$

Table 2. Continuous distributions